CNN-Powered Machine Learning Model for Detection of Common Leaf Diseases in Plants

EAI 421 – Applications of AI & ML in Agriculture

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Outline

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Introduction

Problem Statement

Early diagnosis of common leaf diseases remains a **critical challenge** in modern agriculture, especially for farmers in rural and under-resourced settings.

Key Challenge

Limited access to agricultural experts, costly lab testing, and slow manual inspection methods result in **significant crop losses** every year.

Our Solution

A **CNN-powered web application** that:

- Accepts leaf image input
- Predicts plant diseases in real-time
- Displays preventive measures
- Requires no technical expertise

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Project Objectives

1. Provide an **intuitive web/mobile interface** for farmers to upload leaf photos and get real-time disease forecasts.
2. Incorporate **CNN-based ML models** trained on plant-image datasets to accurately predict disease likelihood.
3. Thoroughly **test and validate** prediction models for high accuracy, sensitivity, and generalization.
4. Deliver **actionable treatment recommendations** including preventive measures and organic remedies.
5. Ensure **scalable and cloud-deployable** architecture accessible from phones and desktops.

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Existing Systems & Their Limitations

Current Approaches

- Expert visual inspection
- Laboratory testing
- Single-crop SVM / Random Forest models
- CNN models targeting one plant species

Key Limitations

1. Limited disease coverage
2. Static input data only
3. Poor data quality / diversity
4. Low model interpretability
5. Lack of field validation

Existing systems lack a **single, unified, user-friendly** *platform covering multiple diseases.*

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Proposed System

Overview

An AI-based web application leveraging **deep learning CNN models** to identify multiple plant leaf diseases from user-uploaded images.

How it works:

1. User uploads a leaf image
2. System validates & preprocesses image
3. CNN model classifies disease
4. Results displayed instantly
5. Preventive tips provided

Key Advantages:

- Multiple disease detection
- Real-time predictions
- No installation required
- Modular & scalable design
- Cloud-ready architecture

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Tools and Technologies Used

Backend & ML

Python	Core language
PyTorch (Torch)	CNN training
TensorFlow/Keras	Model support
Flask	Web framework
OpenCV / PIL	Image processing

Frontend & IDE

HTML/CSS	UI design
VS Code	Development IDE
Jupyter	Notebook support

Data Handling

NumPy	Numerical ops
Pandas	CSV / data mgmt

Dataset

- Kaggle open repositories
- Plant Village Dataset
- Thousands of labeled leaf images

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Hardware & Software Requirements

Hardware Requirements

Component	Spec
Processor	Intel i3 / i5
Hard Disk	50 GB
RAM	Minimum 4 GB

Software Requirements

Requirement	Spec
OS	Windows 8+
Language	Python 3.8+
Framework	Flask, HTML, CSS
Libraries	NumPy, Pandas
IDE	VS Code / Jupyter

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Feasibility Study

Technical

Uses open-source tools (Python, TensorFlow, Flask, OpenCV). Models saved as `.pt` / SavedModel for efficient loading. CNN trained on PlantVillage dataset.

Economic

No licensing fees. Open-source datasets. No specialized hardware required. Cost effective vs. traditional expert consultations.

Operational

Flask provides an intuitive UI. Works on any device – no installation needed. Suitable for rural farming areas with basic internet access.

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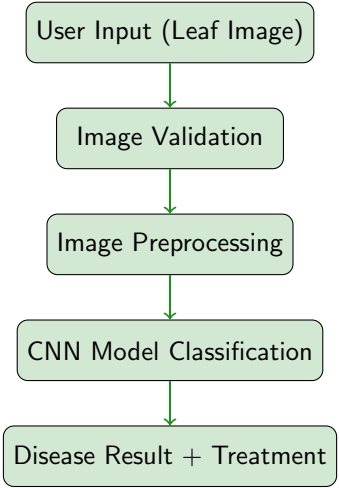
System Design – Architecture



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Functional Requirements

Core Functional Requirements

1. **Leaf Image Upload** – Accept JPG/PNG formats; validate file integrity
2. **Automatic Disease Detection** – No manual selection; CNN classifies automatically
3. **Real-Time Prediction** – Results displayed immediately after upload
4. **Result Display** – Disease name, description, prevention steps, treatment
5. **Multi-Disease Support** – Single platform for various plant disease types

Non-Functional Requirements

Performance $<$ 2 sec response — Usability for non-technical users — Scalability for new diseases

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CNN Model Architecture

Model Pipeline

1. **Input Layer** – Leaf image (resized, normalized)
2. **Convolutional Layers** – Feature extraction (texture, color, lesion patterns)
3. **Pooling Layers** – Dimensionality reduction
4. **Fully Connected Layers** – Classification
5. **Output Layer** – Disease class + confidence score

Training Details

- Framework: **PyTorch**
- Dataset: **Kaggle / PlantVillage**
- Optimizer: **SGD / Adam**
- Saved as: `.pt` model file
- Augmentation: resize, normalize, flip

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Application Screenshots

Entry / Home Page

- Clean, user-friendly dashboard
- Built with Flask + HTML/CSS
- Disease selection interface
- Educational tips for farmers

Results Page

- Detected disease name shown
- Description & severity level
- Treatment recommendations
- Buy product button visible

Upload Page

- Simple file upload input
- Submit button triggers AI engine
- Progress indicator displayed

Contact Page

- Developer contact details
- Email / LinkedIn / Instagram

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System Diagrams

Sequence Diagram Flow

1. User inputs leaf image via UI
2. System validates image
3. Validated image sent to CNN model
4. Model analyzes visual features
5. Disease prediction result returned
6. Result displayed to user

Activity Diagram Steps

1. Upload plant leaf image
2. Receive & validate image
3. Preprocess image data
4. Apply CNN model
5. Predict plant disease
6. Generate prediction report
7. Display disease result to user

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Software Testing Approaches

Testing Strategy

Type	Purpose
Unit Testing	Tests individual components (preprocessing, model loading, prediction function)
Integration Testing	Validates Flask backend + CNN model + HTML/CSS frontend workflow
Acceptance Testing	Verifies system meets real-world farmer/expert needs
System Testing	End-to-end test: image upload → prediction → result display

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Key Test Cases

Test ID	Description	Expected Result	Status
TC01	Valid leaf image upload	Image uploaded successfully	Pass
TC02	Healthy leaf detection	System shows "Healthy"	Pass
TC03	Diseased leaf detection	Disease name displayed	Pass
TC04	Incorrect file format	Error message shown	Pass
TC05	Empty upload submission	Warning message displayed	Pass
TC06	Multiple disease classification	Correct disease identified	Pass
TC07	Prediction accuracy (known data)	Matches expected label	Pass
TC08	Image processing validation	Image accepted by model	Pass
TC09	UI responsiveness	UI adjusts to screen size	Pass
TC10	Disease name accuracy	Correct name returned	Pass
TC11	Medicine recommendation	Correct medicine displayed	Pass
TC12	Buy product button visibility	Button visible on result page	Pass

Conclusion

Summary

The **CNN-powered plant disease detection system** demonstrates a practical, AI-driven approach to modern agricultural support:

- ✓ High accuracy & robustness across varied leaf images and testing scenarios
- ✓ User-friendly interface accessible to non-technical farmers
- ✓ Enables early disease detection, reducing crop loss and pesticide misuse
- ✓ Modular, scalable architecture supporting future disease categories
- ✓ Web-based deployment accessible across all devices
- ✓ Strong foundation for responsible AI use in modern farming

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Future Enhancements

Short-term

- Mobile application development
- Multilingual support for rural farmers
- Expanded disease database
- User authentication & history tracking

Long-term

- IoT sensor & drone integration
- Cloud deployment (AWS/Azure)
- Weather-driven disease prediction
- AI chatbot for farming assistance
- Offline mode for rural areas

Vision

Transform the platform into a **smart farming ecosystem** with real-time monitoring, personalized recommendations, and government agricultural database integration.

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Thank You!

CNN-Powered Machine Learning Model
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